Silas Bambil

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Education

Georgia State University August 2026 to Present
MS Mathematics, concentration in Statistics | Atlanta, Georgia

Georgia State University May 2025
MS Quantitative Risk Analysis and Management | Atlanta, Georgia

Kwame Nkrumah University of Science and Technology July 2020
BSc Actuarial Science | Kumasi, Ghana

Research Interests

Statistical prediction and model evaluation; probability calibration; distribution shift and model adaptation; extreme value theory and tail-risk modeling; Monte Carlo methods; credit risk; statistical and quantitative risk modeling.

Research Experience

Quantifying Catastrophic Fire-Loss Severity Using Peaks-Over-Threshold Extreme Value Theory 2026
Independent research manuscript, unpublished; code and reproducibility archive: Zenodo, DOI 10.5281/zenodo.23253824

Analyzed approximately 1.25 million positive-loss U.S. structure-fire records from the National Fire Incident Reporting System (NFIRS), covering 2016–2025.

Applied Peaks-Over-Threshold extreme value methods and Generalized Pareto models to estimate catastrophic loss severity using Value-at-Risk and Expected Shortfall.

Evaluated robustness through threshold sensitivity, parametric bootstrap, annual tail estimation, leave-one-year-out analysis, event-level aggregation, goodness-of-fit diagnostics, and competing tail models.

Showed that extreme-loss quantiles were comparatively stable across modeling choices, while Expected Shortfall was substantially more sensitive to threshold selection and far-tail uncertainty.

Predictive Stability and Adaptation Under Distribution Shift 2026
Independent research manuscript, unpublished

Studied calibration, discrimination, probability accuracy, and decision thresholds using a loan-default benchmark and controlled Monte Carlo experiments.

Ran 500 replications per condition in R to examine covariate shift with model misspecification, baseline-risk changes, and changes in predictor effects.

Compared intercept recalibration, intercept-and-slope recalibration, importance-weighted refitting, deployment refitting, and threshold revision using independent evaluation samples.

Greenberg Sprint Predictive Enrollment Model 2024
Graduate research project | Georgia State University

Contributed to the development of a predictive model of graduate-student enrollment probabilities using behavioral and demographic data.

Prepared and cleaned applicant data, developed predictor variables, and supported model evaluation for graduate recruitment analysis.

Application of Extreme Value Theory to Fire Insurance 2020
Undergraduate thesis | Kwame Nkrumah University of Science and Technology

Examined extreme value methods for modeling severe fire-insurance losses and tail-risk behavior.

Teaching Experience

Georgia State University August 2026 to Present

Graduate Lab Assistant | Atlanta, Georgia

Teach and support students in College Algebra (MATH 1111) and Support for College Algebra (MATH 0999).

Assist students with mathematical reasoning, problem solving, and foundational algebraic concepts.

Osceola School for the Arts January 2026 to June 2026

AP Mathematics Teacher | Osceola County, Florida

Taught AP Calculus, AP Precalculus, and AP Statistics.

Prepared lessons and assessments, explained mathematical and statistical concepts, and supported students in solving quantitative problems.

Achieved a 70% AP examination pass rate among students taught.

Professional Experience

Georgia State University August 2023 to June 2025

Graduate Assistant | Atlanta, Georgia

Supported predictive enrollment analysis and faculty risk-analysis projects through statistical analysis, research assistance, and data management.

OPH CHEM CONSULT May 2022 to August 2023

Finance Associate | Ghana

Conducted financial research on misappropriation risks and community financial challenges and contributed to analytical reports and briefs.

Ghana Statistical Services September 2020 to August 2021

Research Assistant | Ghana

Contributed research, statistical analysis, and data-management support to population and economic studies.

Technical Skills

Programming and software: R, Python, SAS, Microsoft Excel, and Microsoft Office Suite.

Statistical methods: logistic regression, multiple regression, Monte Carlo simulation, predictive model evaluation, probability calibration, bootstrap methods, data cleaning, and statistical visualization.

Risk and actuarial methods: extreme value theory, Peaks-Over-Threshold modeling, Generalized Pareto models, Value-at-Risk, Expected Shortfall, credit-risk analysis, stress testing, and scenario analysis.

Awards and Certifications

Graduate Merit Scholarship, Department of Mathematics and Statistics, Georgia State University | August 2026

Graduate Merit Scholarship, Robinson College of Business, Georgia State University | August 2023 to May 2025

FINRA Securities Industry Essentials (SIE) Examination | January 2025

FINRA Uniform Securities Agent State Law Examination (Series 63) | August 2025